A

Project Report on

Smart Hospital Appointment and Queue Management System

Submitted for partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

By

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Under the Guidance of

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Certificate

This is to certify that the project entitled " **Smart Hospital Appointment and Queue Management System**" is being submitted by **P. Navya (24K81A05Q1)** in fulfilment of the requirement for the award of degree of **BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING** is recorded of Bonafide work carried out by them. The result embodied have been verified and found satisfactory.

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DECLARATION

I, the student of **"Bachelor of Technology in Department of Computer Science and Engineering"**, session: 2024-2028, **St. Martin's Engineering College, Dhulapally, Kompally, Secunderabad** , here by declare that the work presented in this project work is **Smart Hospital Appointment and Queue Management System** is the outcome of our own Bonafide work and is correct to the best of our knowledge and this work has been undertaken taking care of Engineering Ethics. This result embodied in this project report has not been submitted in any university for award of any degree.

P. NAVYA – 24K81A05Q1

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The satisfaction and euphoria that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose encouragement and guidance have crowded our efforts with success.

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CONTENTS

1. ABSTRACT	**1**
2. INTRODUCTION	**2**
3. LITERATURE SURVEY	**3**
4. SYSTEM REQUIREMENT	**5**
5. FLOW CHART	**6**
6. ALGORITHM	**7**
7. SOURCE CODE	**11**
8. EXPERIMENTAL RESULTS	**12**
9. CONCLUSION	**13**
10.FUTURE ENHANCEMENT	**14**
11.REFERENCES	**15**

1. ABSTRACT

The Smart Hospital Appointment and Queue Management System is a Flutter-based application developed to simplify the process of booking hospital appointments and managing patient queues. In traditional hospital systems, patients often spend a lot of time waiting for consultations, and managing appointments manually can be difficult for hospital staff. This project aims to reduce these problems by providing a simple, user-friendly, and efficient digital solution. The application allows patients to register, log in, view available doctors, check their specializations, book appointments, and view their appointment details. It also provides queue management features that help patients check their queue numbers and monitor their waiting status. Hospital staff can manage appointment details, update queue information, and organize patient visits more efficiently. The user interface is designed using Flutter, which supports the development of attractive and responsive application screens. The system focuses on improving accessibility, reducing unnecessary waiting time, minimizing manual work, and providing a better experience for both patients and hospital staff. Overall, the Smart Hospital Appointment and Queue Management System offers a convenient and organized approach to hospital appointment scheduling and queue management.

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2. INTRODUCTION

The Smart Hospital Appointment and Queue Management System is a software application designed to make hospital appointment booking and patient queue management easier and more efficient. In hospitals, patients often face long waiting times, crowded waiting areas, and difficulties in scheduling appointments with doctors. Managing patient information and appointments manually can also increase the workload of hospital staff. To overcome these challenges, a smart digital system can be used to improve the hospital management process.

This project is developed using Flutter as part of the UI-D Flutter Laboratory. Flutter is a user interface development framework that helps developers create attractive, interactive, and responsive application screens. The application provides a simple interface through which patients can register, log in, view doctors, check their specializations, and book appointments according to their requirements.

The system also includes queue management features that allow patients to view their queue numbers and appointment status. These features help patients understand their position in the queue and enable hospital staff to organize patient visits more effectively. The application aims to reduce confusion, minimize unnecessary waiting, and improve communication between patients and hospital staff.

The main objective of this project is to develop a user-friendly hospital appointment and queue management application that saves time and simplifies the appointment process. By using a digital interface, the system can improve convenience, accessibility, and efficiency in hospital services. In the future, additional features such as appointment reminders, online payments, digital prescriptions, and real-time queue updates can be incorporated to make the application more useful.

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3. LITERATURE SURVEY

The literature survey provides an overview of existing hospital appointment booking systems, patient queue management methods, and mobile application technologies. It helps to understand the limitations of traditional systems and identify the features required to develop the Smart Hospital Appointment and Queue Management System using Flutter.

Traditional Hospital Appointment Systems

Traditional hospital appointment systems mainly depend on manual registration, telephone calls, and direct visits to hospitals. Patients often need to visit the hospital to book appointments and wait for their turn. These methods may lead to overcrowding, longer waiting times, and difficulties in managing patient records. Manual appointment management also increases the workload of hospital staff and may cause scheduling errors.

Online Appointment Booking Systems

Online appointment booking systems allow patients to schedule consultations without visiting the hospital in person. Patients can view doctor details, select suitable appointment times, and access their booking information through a digital platform. These systems improve convenience and reduce the need for manual appointment registration. However, some systems may not provide proper queue tracking or real-time updates about consultation status.

Patient Queue Management Systems

Queue management systems are designed to organize patients according to their registration numbers or appointment schedules. These systems help hospital staff manage patient flow and reduce confusion in waiting areas. Digital queue systems may provide token numbers and display the current queue status. However, basic systems may lack user-friendly interfaces and convenient access to queue information through mobile applications.

Proposed Smart Hospital Appointment and Queue Management System

The proposed system combines appointment booking and queue management in a single Flutter-based application. It allows patients to register, log in, view doctor information, book appointments, and check their appointment and queue status. It also provides a structured interface for managing appointment details and patient queues. Compared with traditional methods, the proposed system aims to improve convenience, reduce manual work, and make hospital visits more organized.

4. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Hard Disk: Minimum 10 GB of free space
- System: Desktop computer or laptop
- Display: Standard monitor with suitable screen resolution

Software Requirements

- Operating System: Windows 10 or Windows 11
- Programming Language: Dart
- Framework: Flutter
- IDE: Visual Studio Code or Android Studio
- SDK: Flutter SDK and Android SDK
- UI Design: Flutter Material Design widgets
- Testing Platform: Android Emulator or physical Android device
- Database: Firebase or SQLite, depending on the implementation
- Version Control: Git (optional)

Functional Requirements

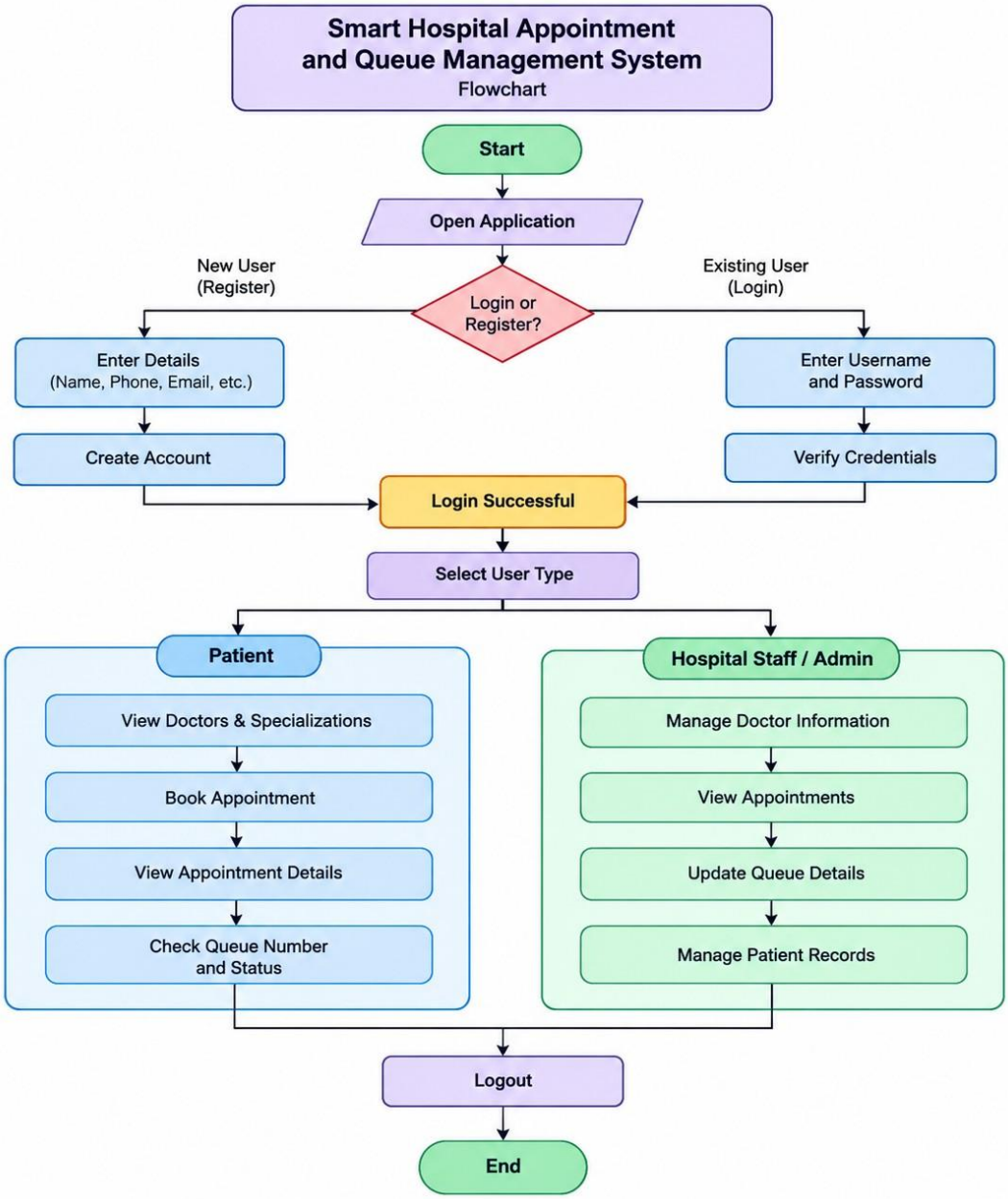
The application should allow users to register and log in, view doctor details, check doctor specializations, book appointments, and view appointment information. It should also provide queue numbers and appointment status. An administrator or hospital staff member should be able to manage doctor information, appointments, and patient queue details, depending on the implemented features.

Non-Functional Requirements

The application should have a simple, attractive, and user-friendly interface. It should respond quickly to user actions and provide reliable appointment information. The design should be responsive acrossdifferent screen sizes, and patient information should be protected through appropriate security

Measures.

5. FLOW CHART



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6. ALGORITHM

The algorithm describes the step-by-step process of the Smart Hospital Appointment and Queue Management System developed using Flutter. It explains how users register or log in, book appointments, and manage patient queues.

Steps:

1. Start the application.
2. Display the home screen with login and registration options.
3. If the user is new, enter the required details and create an account.
4. If the user already has an account, enter the username and password.
5. Verify the login details and allow access to the application.
6. Display the available options based on the user type.
7. If the user is a patient:
    - View available doctors and their specializations.
    - Select a doctor and book an appointment.
    - Display the appointment details.
    - Generate or display the queue number and appointment status.
8. If the user is hospital staff or an administrator:
    - Manage doctor information.
    - View and manage appointments.
    - Update queue details.
    - Manage patient records.
9. Display the updated appointment and queue information.
10. Allow the user to log out of the application.
11. Stop the process.

7. SOURCE CODE

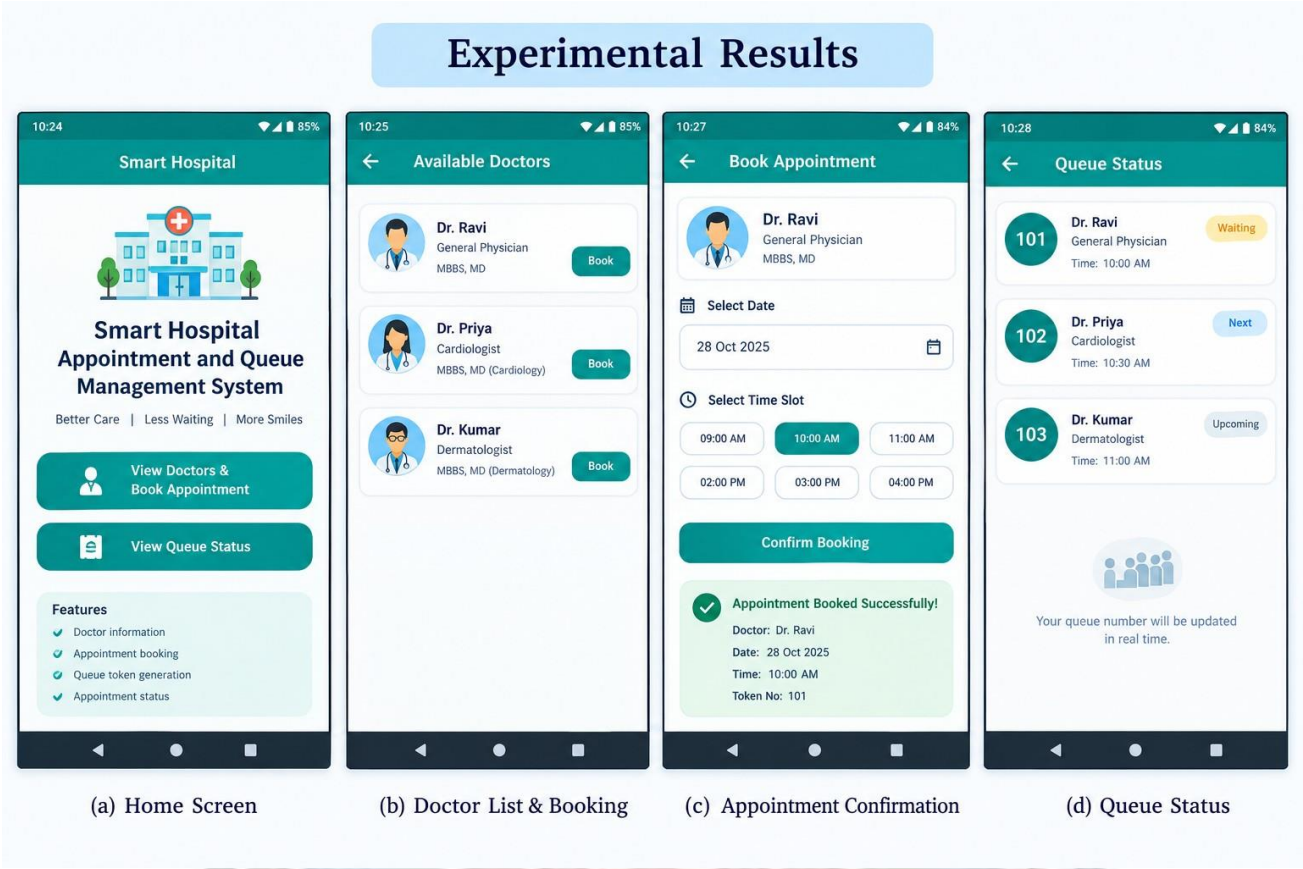
```
import 'package:flutter/material.dart';
void main() {
  runApp(const HospitalApp());
}
class HospitalApp extends StatelessWidget {
  const HospitalApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Smart Hospital',
      theme: ThemeData(
        primarySwatch: Colors.teal,
      ),
      home: const HomePage(),
    );
  }
}
class HomePage extends StatefulWidget {
  const HomePage({super.key});
  @override
  State<HomePage> createState() => _HomePageState();
}
class _HomePageState extends State<HomePage> {
  final List<String> doctors = [
    'Dr. Ravi - General Physician',
    'Dr. Priya - Cardiologist',
    'Dr. Kumar - Dermatologist',
  ];
  final List<String> appointments = [];
  int token = 100;
  void bookAppointment(String doctor) {
    setState(() {
      token++;
```

```
    appointments.add('$doctor | Token No: $token');
  });
  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(content: Text('Appointment booked. Token: $token')),
  );
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Smart Hospital'),
      centerTitle: true,
    ),
    body: Center(
      child: Padding(
        padding: const EdgeInsets.all(20),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            const Icon(Icons.local_hospital,
                size: 80, color: Colors.teal),
            const Text(
              'Smart Hospital Appointment\n'
              'and Queue Management System',
              textAlign: TextAlign.center,
              style: TextStyle(
                fontSize: 22,
                fontWeight: FontWeight.bold,
              ),
            ),
            const SizedBox(height: 25),
            ElevatedButton(
              onPressed: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(
```

```
                builder: (_) => DoctorPage(
                  doctors: doctors,
                  onBook: bookAppointment,
                ),
              ),
            );
          },
          child: const Text('View Doctors & Book Appointment'),
        ),
        ElevatedButton(
          onPressed: () {
            Navigator.push(
              context,
              MaterialPageRoute(
                builder: (_) =>
                    QueuePage(appointments: appointments),
              ),
            );
          },
          child: const Text('View Queue Status'),
        ),
      ],
    ), ), ), );
  }
}
class DoctorPage extends StatelessWidget {
  final List<String> doctors;
  final Function(String) onBook;
  const DoctorPage({
    super.key,
    required this.doctors,
    required this.onBook,
  });
  @override
  Widget build(BuildContext context) {
    return Scaffold(
```

```
      appBar: AppBar(title: const Text('Available Doctors')),
      body: ListView.builder(
        itemCount: doctors.length,
        itemBuilder: (context, index) {
          return Card(
            child: ListTile(
              leading: const Icon(Icons.person),
              title: Text(doctors[index]),
              trailing: IconButton(
                icon: const Icon(Icons.book_online),
                onPressed: () => onBook(doctors[index]),
              ), ), );       },
      ),
    );
  }
}
class QueuePage extends StatelessWidget {
  final List<String> appointments;
  const QueuePage({
    super.key,
    required this.appointments,
  });
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Queue Status')),
      body: appointments.isEmpty
          ? const Center(child: Text('No appointments booked yet'))
          : ListView.builder(
              itemCount: appointments.length,
              itemBuilder: (context, index) {
                return ListTile(
                  leading: const Icon(Icons.confirmation_number),
                  title: Text('Appointment ${index + 1}'),
                  subtitle: Text(appointments[index]),
                );}),);}}
```

8. EXPERIMENTAL RESULTS



(a) Home Screen (b) Doctor List & Booking (c) Appointment Confirmation (d) Queue Status

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9. CONCLUSION

The Smart Hospital Appointment and Queue Management System is a Flutter-based application developed to make hospital appointment scheduling and patient queue management easier, faster, and more organized. The main purpose of this project is to reduce the difficulties faced by patients during hospital visits, such as long waiting times, overcrowded waiting areas, and problems with manual appointment booking. The application provides a simple and interactive user interface that allows patients to view available doctors, check their specializations, book appointments, and view their appointment details.

The system also includes queue management features that help patients check their token numbers and appointment status. By organizing appointment and queue information digitally, the application can help reduce confusion and improve the overall hospital experience. It also helps hospital staff manage patient appointments and queue details more efficiently. The use of Flutter makes it possible to develop an attractive, responsive, and user-friendly interface that can be adapted to different screen sizes.

This project provides practical experience in mobile application development, UI design, navigation between screens, event handling, and state management using Flutter and Dart. It demonstrates how technology can be used to solve common problems in hospital management and improve communication between patients and hospital staff.

In conclusion, the Smart Hospital Appointment and Queue Management System offers a convenient digital approach to hospital appointment scheduling and queue organization. It aims to save time, reduce manual effort, improve accessibility, and provide a better experience for patients and hospital staff. With further development and the addition of advanced features, the application has the potential to become a more complete and efficient hospital management solution.

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10. FUTURE ENHANCEMENT

The Smart Hospital Appointment and Queue Management System can be improved in the future by adding advanced features to make hospital services more efficient, reliable, and convenient for patients and hospital staff. The current application focuses on the user interface, doctor details, appointment booking, and queue status. Several additional features can be implemented to enhance its functionality.

In the future, the application can be connected to a database such as Firebase or MySQL to store patient details, doctor information, and appointment records securely. User authentication can be implemented to provide separate login access for patients, doctors, and administrators. Real-time queue tracking can also be introduced so that patients can check their current queue position and receive updates about their expected waiting time.

Automatic notifications and reminders can be added to inform patients about upcoming appointments, booking confirmations, cancellations, and changes in doctor availability. Online payment facilities can be integrated to allow patients to pay consultation fees through the application. The system can also include features such as online consultation, digital prescriptions, medical report management, and appointment rescheduling.

Furthermore, artificial intelligence can be used to provide basic guidance for selecting the appropriate medical department based on the patient's reported needs. The application can also be enhanced with multilingual support, accessibility features, and an improved dashboard for hospital administrators. These improvements can make the system more useful for different types of users.

Overall, future enhancements will help transform the application into a comprehensive hospital management platform that improves appointment scheduling, reduces unnecessary waiting, and provides a smoother experience for patients and hospital staff.

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